Supply Chain Transformation Analysis:

TSMC's Geographic Expansion and Supply Chain Sovereignty

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Executive Summary

The high geographic concentration of the Advanced semiconductor manufacturing of semiconductor in Taiwan by TSMC is a significant weakness in the international supply chain. However, TSMC's geographic expansion sits at an interesting intersection: genuine supply chain challenges meet strategic considerations around what defense analysts call the "Silicon Shield," where Taiwan's security arguably depends on the world's reliance on its semiconductor manufacturing (Miller, 2022).

In the light of this plight of the Arizona plant, 300 percent cost overruns, delays in production, and cultural collisions I see not only genuine working hurdles but also results that are point-blank illustrations of the benefits of Taiwan. My assessment has shown that there is more than just strategic resistance: TSMC experiences actual ecosystem dependencies, tacit knowledge transfer issues and coordination complexities that coincidentally fit the strategic interests in ensuring that Taiwan can remain central. It is difficult to tell whether these challenges are part and parcel of complexity or strategic calculation, but comprehending the way they influence the positioning of TSMC is more important than knowing whether they are deliberate or not.

The essence of the question that the analysis is trying to answer is as follows: can semiconductor manufacturing be effectively distributed keeping efficiency and innovation: my analysis finds that it is, but heavily qualified. Distribution may develop operational functionality, namely Arizona will make state-of-the-art chips, but it cannot reach Taiwan-level efficiency over 5-10 years time, the dependence of real ecosystems, and tacit knowledge demands. The 36 point gap in SCOR, the 30 percent cost premium and the 3-5 year convergence period of yield reveal that during long transitions, there are significant sacrifices in efficiency. Innovation is even more problematic: the innovative development is associated with the focused expertise that is disintegrated by distribution. The capacity distribution facilitated by TSMC with innovation centralization maintenance is due to the staged deployment of technology- Arizona is always 1 or 2 generations behind Taiwan. Distribution, hence, is effective in imitating the mature

manufacturing capacities but fails to imitate the Taiwanese operational efficiency or leadership of innovations with a horizon of policy-related periods of 5-10 years.

This analysis examines TSMC's expansion through the SCOR framework, technology assessment, risk management, and performance metrics to understand how genuine supply chain challenges create both operational problems and strategic benefits.

1. SCOR Framework Analysis: Real Complexity, Strategic Alignment

SCOR Performance Comparison

The table below shows how TSMC's operations differ across SCOR dimensions, revealing the capability gap between Taiwan and Arizona:

Table 1: TSMC SCOR Performance Scorecard Comparison

SCOR Process	Taiwan Score	Arizona Score	Key Gap Drivers	Strategic Implication
Plan	19/20	12/20	Time zone coordination; workforce planning; technology staging	Taiwan has innovation leadership on a staged deployment basis.
Source	20/20	11/20	Supplier maturity (30% vs. 100% local); material qualification; lead times	Proves ecosystem requires generational development
Make	19/20	13/20	Yield gap (90% vs. 75%); tacit knowledge transfer; cultural adaptation	Manufacturing excellence depends on non-replicable factors
Deliver	18/20	14/20	Market proximity Asian markets; regulatory reality; global supply chain reality.	Limited geographic advantages from distribution
Enable	20/20	10/20	Talent pipeline; institutional knowledge; autonomous capabilities	Creates legitimate long-term dependencies
OVERALL SCORE	**96/100**	**60/100**	36-point capability gap	Compromise between the real world of operations and the strategic advantage.

SCOR Performance Calculation: Overall scores represent the sum of individual process ratings. Taiwan's 96/100 derives from Plan (19) + Source (20) + Make (19) + Deliver (18) + Enable (20) = 96 points, demonstrating mature, integrated operations across all dimensions. Arizona's 60/100 derives from Plan (12) + Source (11) + Make (13) + Deliver (14) + Enable (10) = 60 points, reflecting developing capabilities with significant deficiencies. The 36-point gap indicates a 37.5% capability deficit, concentrated primarily in Enable (50% capability), Source (55% capability), and Plan (60% capability) processes.

1.1. Plan: Complexity That Serves Multiple Purposes

The Arizona planning of TSMC has real coordination issues, such as capacity coordination between continents, allocation of limited equipment with lead times of 18 months, and negotiating varied regulatory landscapes (Chopra and Meindl, 2023). The choice of 3-nanometer production in Arizona and Taiwan developing to 2-nanometer technology is a decision that is partly a practical risk management. The application of state-of-the-art processes within new plants with inexperienced work forces will be associated with significant technical risk that will cost billions to the investment.

However, this staged approach conveniently preserves Taiwan's technological leadership. Whether TSMC could accelerate Arizona's roadmap remains uncertain, but the company faces little incentive to find out when practical challenges provide legitimate justification for approaches that serve strategic purposes.

The talent challenge seems authentic to me. Each fab needs around 2,000 PhD-level engineers, and Taiwan's universities produce thousands annually through decades of industry partnership (Bloomberg, 2023). This kind of infrastructure is lacking at Arizona- this is not a strategic fabrication, this is an actual state of affairs. However, the focus on this shortage is useful in controlling expectations regarding the timeline of Arizona to reach parity and these longer timelines continue to favor Taiwan.

This is an explicit technology deployment plan that is granted by the Taiwanese authorities. In an interview with the Wall Street Journal, Taiwan's Representative to the United States, Alexander Yui, confirmed that TSMC will manufacture mainstream semiconductors in the U.S. and other global locations, but the most advanced cutting-edge chips will be produced first in Taiwan before eventually deploying to other facilities (Bade, 2024). This is an official statement, and it confirms what operational evidence has been saying and that is, technology staging has not been just an implementation limitation but is purposeful strategic policy that Taiwan is always at the frontline of innovation. The fact that Phoenix and other foreign fabs will be introduced to modern technologies later officialises the generational divide that will remain between Taiwan and other states no matter how successful the distribution facilities will be in the end.

1.2. Source: Genuine Ecosystem Dependencies

The sourcing issues encountered by Arizona are a result of actual complexity of the supply chain. Semiconductor production demands more than 300 special materials with parts-per-trillion purity which a few suppliers around the world are capable of supplying (Semiconductor Industry Association, 2022). It has been growing through decades in Taiwan with a supplier ecosystem where TSMC and its suppliers co-invested in capabilities and specifications; this is a true competitive advantage due to geographic proximity and accumulated collaboration (Porter, 1998).

These were not artificial obstacles but genuine quality deficiencies when Arizona suppliers did not pass the purity tests by TSMC on industrial gases (Wall Street Journal, 2023). Years of work produce only 30 percent of the materials locally in Arizona since building supplier strengths takes years and technical expertise to make them do things faster than arbitrary (Nikkei Asia, 2024).

However, TSMC faces discretion around supplier development investment intensity. Greater aggressiveness in investment might enhance the development of the Arizona ecosystem faster and decrease the relative advantages of Taiwan. The level of the current investments signifies good will and results are limited by the real technical limiting factors that benefit the strategic benefits conveniently.

1.3. Make: Where Operational Reality Meets Strategic Value

The manufacturing transfer issues are a good example of what I consider the most evident point of real complexity and strategic advantage. Semiconductor manufacturing is a process containing more than 1,000 process steps in which nanometer accuracy is required and high yields are highly reliant on tacit knowledge-experiential knowledge that engineers acquire over a period of hands- on experiences (Nonaka and Takeuchi, 1995).

The study establishes the fact that tacit knowledge is difficult to capture and transmit and more so across cultures. When 500 Taiwanese engineers taught Americans, there were real blocks due to differences in culture through work intensity and the hierarchy structure of the company (Reuters, 2023). The American demands of arranged schedules face Taiwanese demands of flexibility and deference of a superior--they are genuine differences in the norms of labor markets, and these are not cruel barriers it creates.

The estimated 3-5 years of Arizona to reach Taiwan-like yields indicate the actual learning needs as well as convenient strategic deliverable results (Nikkei Asia, 2024). Traditionally new sites take years before streamlining processes - that is the reality of operations in any manufacturing sector. Yet highlighting these difficulties allows the management of expectations and the schedule makes Taiwan stay several generations ahead as technology keeps evolving.

The issue of Arizona chip quality among customers is an actual inquiry on the possibility of the distributed operations in Arizona to comply with the standards of Taiwan (Bloomberg, 2024).

Such fears are based on the actual manufacturing complexity rather than the strategic discourse that TSMC has been selling--but they are happy to support those ideas that Taiwan manufacturing is still an ecstasy of perfection.

1.4. Deliver and Enable: Limited Benefits, Authentic Gaps

Geographic diversification provides less delivery value than expected due to supply chain realities. iPhones contain components from dozens of global suppliers co-locating one supplier offers limited benefit if others still source from Asia (Christopher & Peck, 2004). Also, U.S.-made chips are exported with restrictions that place compliance burdens on Taiwan, something that the single regulatory environment of Taiwan avoids (Semiconductor Industry Association, 2023).

The enabling infrastructure gaps are perhaps the most natural barriers that I have identified. TSMC's corporate universities and training systems developed over 40 years cannot be quickly recreated (Garvin, 2000). The educational system of Taiwan developed the semiconductor engineering programs due to the policy of the government and industry cooperation over decades. The state of Arizona does not have such infrastructure-to mitigate it, generations long dedication that lies outside the control of any single corporation would be needed.

2. Technology Assessment:

2.1. Digital Twins: Genuine Tool with Natural Limits

Digital twins obtain valid operation assistance- Arizona engineers can model Taiwan parameters and process interaction (Grieves and Vickers, 2017). The limitation of simulations has however well been documented in the literature. Such instances of prediction being different from real behavior indicate genuine incompleteness of codified knowledge (Tao et al., 2019).

I believe that the technology can be used to achieve two goals: both to offer some real value and at the same time illustrate how much tacit knowledge cannot be captured by digital systems. Taiwan engineers can recognize the issues in Arizona due to the digital twins, but the explanation of underlying causes will involve the experiential knowledge that cannot be transferred to models. This constraint is inherent to the nature of knowledge, not withholding strategy there are things that knowledge really can not formulate.

2.2. AI: Automating Patterns, Not Judgment

AI use in quality control has the advantage of recognizing patterns, sensor data is analyzed with AI finding out patterns to predict defects (Cioffi et al., 2020). This genuinely augments manufacturing capabilities. AI is also used to aid in the planning processes of TSMC, with the capability to analyze dealer orders, economic indicators, and seasonal trends to forecast the semiconductor demand among vehicle programs (Chui et al., 2022). Nevertheless, AI fails in new scenarios that it has not trained on and does not provide reasons as to how predictions

came to be. One can also find the same drawbacks in these forecasting models, whereby the market conditions change unpredictably beyond the training data set, whereby human judgment is needed to extrapolate the forecast reliability in unstable markets.

The challenge training models that were developed by Arizona first using data in Taiwan are a reflection of actual technical obstacles, the different equipment configurations and material properties cause distribution changes that reduce model performance. These technical facts imply that AI helps to minimize, but not eradicate reliance on human experience, especially in adaptive problem-solving when one faces novel conditions. Taiwan engineers have causal knowledge about the manufacturing processes which allows them to respond to new issues effectively whereas AI offers pattern recognition but no understanding (Brynjolfsson and McAfee, 2022). This authentic ability disparity conserves benefits of accrued human experience.

2.3. Blockchain: Transparency Revealing Dependencies

The adoption of blockchain technology by TSMC satisfies the requirement in supply chain transparency needs and also reveals interesting strategic implications. The blockchain provides material provenance, quality certifications, and handling conditions that are an immutable and complex semiconductor supply chain consisting of 300 + materials supplied by suppliers all around the world (Pournader et al., 2020). The technology has important functions in preventing counterfeits, which is of great concern in semiconductor markets that feature fake chips that can affect vehicle safety and reliability (Saberi et al., 2019).

Nevertheless, the transparency blockchain offers also the records of constant reliance of Arizona on Taiwan-based materials, Taiwan-tested suppliers, and Taiwan-made specifications. The record of each blockchain transaction discloses the extent to which Arizona is dependent on the Taiwan ecosystem -70% of materials continue to be imported into the country even when it has spent years establishing suppliers locally. This visibility is not only used to assure quality but also portrays the extent to which Arizona is still incorporated into the Taiwanese supply chain instead of actual autonomy in its operations (Kouhizadeh & Sarkis, 2018).

It also affects intellectual property protection, with access controls based on blockchain monitoring the access of sensitive technical data, which leaves audit trails, preventing the transfer of knowledge to unauthorized parties (Wang et al., 2019). These security remedies represent real security issues with the diffusion of technology based on geographic expansion, however, it also constrained the capacity of Arizona to build up completely autonomous capabilities. Without Taiwan consent, engineers lack effective access to full technical documentation and Taiwan has maintained a monopoly of access to key knowledge despite geographically dispersing operations.

Implementation of blockchain therefore has two effects: the supply chain becomes more visible and secure (operational gains) and relationships and restrictions to autonomy are recorded (strategic gains). The technology does not allow Arizona to be independent but provides clear evidence of continued Taiwan-centricity in the global TSMC operations (Cole et al., 2019).

2.4. IoT and Real-Time Monitoring: Remote Control Infrastructure

The IoT sensors will give TSMC more visibility than ever in its manufacturing operations distributed throughout the world to follow equipment performance, environmental conditions, and material flows across geographic boundary lines. RFID tags and GPS trackers are used to track semiconductor inventory levels and component locations between manufacturing sites and final assembly sites within logistics networks, so that real-time stock positions can be observed automatically and manual inventory counts can be performed periodically (Atzori et al., 2017).

The IoT technology is not only used to monitor but also allow predictive maintenance. The patterns of degradation of the equipment identified by machine learning algorithms examining the data provided by the IoT sensors monitoring the vibration, temperature, and performance indicators allow predicting the imminent failures and thus repairing the equipment before it goes out of service, reducing any downtime (Wan et al., 2018). In the case of semiconductor manufacturing equipment, which is a billion-dollar investment per plant, this predictive potential is of significant value to avoiding accidental interruptions in production (Zhong et al., 2017).

Nevertheless, the dynamics surrounding operational autonomy are interesting due to the influence of IoT implementation. The real-time monitoring will allow Taiwan specialists to monitor the Arizona equipment activity on a regular basis and at a distance, and thus identify unusual activity and activate interventions prior to the issues impacting quality. This distance supervisory feature came in especially handy when Arizona was ramp-up, as the lack of familiarity to the equipment and inexperience of the operators heightened the possibility of parameter drift and subsequent defects. The remote monitoring and intervention capability of Taiwan made up for the low experienced engineering depth that was found in Arizona (Ben-Daya et al., 2019).

However, this very ability institutionalizes the reliance of Arizona on Taiwan experience instead of the establishment of local independent potential. The IoT system establishes technological systems that demand ongoing Taiwan presence to interpret and make decisions, and allows Taiwan to maintain its operational authority over Arizona instead of allowing Arizona to be independent (Xu et al., 2018).This represents the paradox of enabling technologies: IoT genuinely supports Arizona operations while the support structure embeds rather than reduces concentration of expertise in Taiwan. The technology allows the geographic decentralization of manufacturing operations but centralized manufacturing intelligence and decision-making.

2.5. The Core Insight: Human Systems Trump Hardware

So far, the most interesting thing that has struck me is that TSMC has its sustainable advantage in the human knowledge systems as opposed to equipment. The extreme ultraviolet (EUV) lithography devices of ASML in the Netherlands can be the most vivid illustration of the role and impact paradox of technology. These 100,000-part, 100,000-part machines, made by 800 suppliers in the world, are the best in the manufacturing technology, but the ownership does not give any automatic competitive advantage (Financial Times, 2023). Intel, Samsung, and TSMC all purchase identical EUV equipment from the same monopoly supplier. The key difference is

not in the ownership of hardware, but the built experience of using these exceptionally complicated tools as optimally as possible. Over decades the engineers of Taiwan have worked on several generations of lithography devices and have gained a deep feeling of the delicate behavior of machines, interactions of processes and methods of troubleshooting that allow high yields (Fuller, 2022). The engineers of the first generation of Arizona that use the same EUV machines get lower yields in the early days since they do not have this experience in their hands- that is to say that technology actually exerts its effects not based on the specifications of the hardware but on the human knowledge systems.

EUV machines, AI algorithms, digital twins, blockchain platforms, and IoT sensors are highly advanced technologies that can be bought or created by competitors.Intel owns similar equipment; Samsung develops comparable AI systems (Khan et al., 2021).The accrued tacit knowledge of how to leverage these technologies, or experiential expertise on the behavior of equipment and process interactions that concentrates in Taiwan over decades of operation is what gives competitive advantage (Nonaka and Takeuchi, 2019).

This is real supply chain reality, which is validated in knowledge management research literature: tacit expertise takes long periods of time to build, and it does not transfer readily across contexts. Unreasonable obstacles are not found in TSMC because of the authentic learning accumulation (Oliveira et al., 2021). But the real benefit poses circumstances wherein retaining knowledge concentration generates instances of operational excellence and strategic positioning, the motivations are hard to unravel.

3. Risk Management: Navigating Genuine Vulnerabilities and Strategic Interests

3.1. The Stated Risk: A Global Chokepoint

The first factor that has led to TSMC geographical expansion is the universally recognized threat of its excessive concentration in Taiwan. The TSMC manufactures 92 percent of the most advanced chips in the world, as is mentioned in the case study, and it is a key chokepoint to the global economy. The worry is that in case an invasion or military blockade by the Chinese killed or crippled these fabs, it would permanently discontinue the semiconductor chain around the globe, and that economists estimated it would cost the world trillions of dollars (Miller, 2022).

This possible interference is a fundamental national security issue to the Western powers.Therefore, pressure is being exerted by the United States and the European Union on TSMC to expand its manufacturing presence.They are putting huge subsidies such as the U.S. CHIPs Act to attract TSMC to construct fabs in their soil and ensure local supply. This is the immediate risk that is perceived by the people and which TSMC is getting paid to fix.

3.2. The Risk Paradox: Multiple Conflicting Objectives

It is a simple demand of diversification, but it is what I perceive to be the fundamental paradox of the TSMC strategy. Although it helps reduce the global supply chain risk to its customers, it has a direct threat of undermining Taiwanese Silicon Shield. It has become real complex risk landscapes that demand refined navigation by TSMC; the Taiwan-China tensions, supply chain complexity, the possibility of a loss of competitive advantage, and the customer risks of quality expectation (Tang, 2006). These are real business problems and not artificial complications.

The concept of the Silicon Shield is based on the actual geopolitics. The economic weight of Taiwan in the form of semiconductors has provided an actual deterrence that interrupting production would cause economic impacts on the whole world (Miller, 2022). The U.S. authorities admit that semiconductor reliance affects defense policy, and some analysts think that America would defend Taiwan to gain an access to chips, instead of being fully ideologically devoted to democracy (Semiconductor Industry Association, 2023). This is a true geopolitical reality which affects the risk assessment.

Nevertheless, TSMC should overcome the tension between mitigation measures to risk. Geographic diversification effectively replicating Taiwan capabilities would decrease geopolitical concentration risk, but may also decrease the deterrence value- one mitigation strategy is in conflict with another. This is a true strategic dilemma to which there is no apparent best alternative answer.

3.3. Supply Chain Resilience Assessment: Current State and Future Capability

The present resilience profile of TSMC is highly susceptible with extreme geographical concentration. Taiwan is a primary supplier of advanced semiconductor with over 92 percent of the total output in the region, and any disruption would result in immediate shortages worldwide with no alternative options (Miller, 2022). Just in Time inventory practices by the industry imply that the majority of their customers are holding a 2-3 months stock in the chips, which results in the establishment of very thin buffers before the effects of the production stand-off permeate the industries.

Geographic diversification theoretically increases resilience by having redundant capacity that is immune to Taiwan-specific disruptions. But the State of Arizona has a 60/100 SCOR capability score which shows there is little substitution potential - in case of Taiwan operations, Arizona would be in a position to have only 25-30 percent effective production in view of yield shortages, capacity restriction and reliance on imported materials. This partial resiliency is better than zero diversification but is still insufficient to provide any meaningful supply security over 5-10 year time frames required by policy and market forces (Sheffi & Rice, 2005).

Recovery time analysis displays other constraints. Arizona is unable to ramp up production quickly to offset Taiwan disruption-the 3-5 year window to attain Taiwan-equilivalent yields would imply that recovery facilities are constrained. The problem of equipment is exacerbated: the lead

times of ASML EUV machines are 18 months, and there are few installed worldwide. The lack of talent would create a bottleneck since thousands of specialized engineers would not be available in the US labor markets during crisis periods to scale (Fuller, 2022).

The efficacy-resilience trade-off brings about strategic tensions. To retain enough Arizona capacity to respond to any Taiwan backup, operation plus investment must continue even in normal times with high costs, to the tune of about 6-8 billion/per year in fixed overhead, to depreciation, and untapped talent. TSMC has to trade these resilience insurance premiums and probability-weighted Taiwan disruption losses and find hard choices of capital allocation between resilience and innovation expenditures and shareholder earnings.

3.4. Geopolitical Risk: Real Threat, Complex Mitigation

The risk posed by the geopolitical issues between the two countries on the issue of Taiwan is not a hypothetical one but a real threat. The escalation of tensions between China and Taiwan poses real-life situations when the supply of semiconductors may be lost (Allison, 2017). In theory geographic diversification can counteract this by spreading production among nations whose security circumstances are independent - this is valid risk management reasoning which is buttressed by supply chain resilience studies (Sheffi & Rice, 2005).

Nevertheless, the effectiveness of mitigation is vital on whether Arizona is capable of becoming Taiwan-equivalent. In case Arizona is 70 percent efficient or is producing older-generation chip, customers would rationally choose to buy Taiwan production when Arizona is functioning normally, and Arizona would become a costly insurance that may not be utilized at all. Diversification will only bring significant risk reduction in Arizona in case it can be compared to Taiwan and its capabilities but that will bring Taiwan to a strategic niche that she can no longer occupy.

TSMC is therefore experiencing real tensions in which the full diversification success is counterposed to the strategic value of Taiwan. The company can legitimately emphasize operational challenges limiting Arizona's development pace while these challenges conveniently preserve Taiwan's relative advantages. The high cost of Arizona construction at 300 percent shows that geographic diversification is a very expensive endeavor and justifies the idea that defense of Taiwan is a less expensive approach than trying to substitute its functions (Reuters, 2024).

The issue of financial risks adds strategic strains in geographic expansion. The Arizona investment of 40 billion dollars constitutes about a quarter of the total market capitalization of TSMC, which poses significant risks of capital allocation in case the facilities will not meet the operational goals or strategic intentions (Reuters, 2024). This is in contrast with Taiwan investments that have shown returns over time, Arizona investments have a lot of uncertainty regarding the possibility of making a profit in the future since there are consistent cost premiums, yield deficits and an uncertain customer demand of the US produced chips at a higher price. The three-fold cost increase compared to Taiwan-equivalent capacity indicates that Arizona will never come up with reasonable returns on investments under regular market

conditions and it will have to be still subsidized by the government to make operations economically worthwhile.

Sunk cost has extra financial hazards where an initial investment exerts strain on further investments irrespective of the declining strategic value. With its investment of $40 billion in Arizona construction and equipment, TSMC is politically and reputational pressured to show success by further investing in the development of suppliers, hiring of talents, and transfer of technology- although rational analysis may indicate that there is insufficient returns to these further investments. This sunk cost trap may result in the increase of commitments to include 60-80 billion across the lifetime of the facilities, much greater than the original estimates and may limit the capital available to invest in innovation in Taiwan or other strategic areas (Kahneman and Tversky, 1979).

3.5. Technology Transfer and Supplier Risks: Authentic Tensions

Tensions of technology transfer are natural products of the expectations of the stakeholders and the competition. The customers and the governments need to know so that they can develop their own abilities; the competitive needs demand that the proprietary expertise that is part of the advantage should be preserved (Kogut and Zander, 1992). The selective transfer strategy in TSMC, i.e. codifiable knowledge sharing and tacit expertise protection, is not a case of unconventional strategic manipulation, but the common practice of management in managing joint ventures and technology licensing.

Such cultural barriers to knowledge transfer are real instances of cross-cultural differences which are always a fact in any organizational research and which are always recognized by the research to be real barriers that take a lot of effort to overcome (Hofstede, 2001). These cultural differences were not made by TSMC intentionally to live up to a strategy; they are actual social processes which occur coincidentally to satisfy strategic interests, by curtailing the completeness of knowledge transfer. It is not yet clear whether TSMC might deal aggressively with these barriers by doing things differently.

The challenges around supplier development are genuine complexity of the ecosystem that has been found out over decades of research on industrial clusters - these ecosystems evolve over time through accrual investment and shared learning (Porter, 1998). In case Arizona cannot identify local suppliers it is based on actual capability gaps where the potential suppliers do not have the experience, equipment, and technical capability of meeting the TSMC requirements. The level of investment that TSMC has placed in supplier development schemes, however, is a discretionary decision in which an aggressive strategy might speed up the process of capability building at the expense of Taiwan owning less advantages of the ecosystem.

3.6. Single Points of Failure and Systemic Vulnerabilities

Geographic diversification will deal with Taiwan concentration risk but will not be able to eradicate many single points of failure with all TSMC facilities irrespective of the location. The narrowest systemic weakness is the monopoly of the extreme ultraviolet lithography systems of the equipment; the sector as a whole relies on EUV equipment sold by a single supplier in the Netherlands (Financial Times, 2023). The machines cost 18-month lead times and have to be maintained with a continuous technical support of ASML. The failure of any ASML system would also impact all TSMC sites and in this case geographic diversification would be pointless with this essential reliance.

Suppliers of ultra-pure materials are also other single points of failures. The manufacture of semiconductors involves chemicals and gases which are purified to parts-per-trillion level and are only available to a few suppliers in the world. A large number of the critical materials are a single-source with proprietary purification methods- a disruption would cause a material shortage at the same time in all facilities without taking into account geographical location (Semiconductor Industry Association, 2022). The case of Arizona in trying to source local chemical suppliers out of redundancy illustrates that it is not an easy task hence there is likelihood that such dependencies will still continue.

The concentration of talent exposes not so evident but equally serious weaknesses. The worldwide scarcity of PhD qualified semiconductor engineers implies that the facilities of TSMC are competing over the few available talent pools and not independent labor markets. The semiconductor experience in Taiwan is a hundred years of human capital that cannot be easily duplicated in other locations, the issue of geopolitical tensions that would lead to engineering talent flow would impact all the plants that rely on the Taiwan knowledge base in one way or another (McKinsey & Company, 2023).

Modern systemic weaknesses are found in information technology infrastructure. The distributed operations in TSMC rely on data networks, cloud computing and communication system to coordinate and monitor their operations remotely. All facilities might be jeopardized at the same time by cyber attacks, ransomware, or by hacking sponsored by the state. IoT sensors and AI systems that allow oversight of Taiwan remotely establish attack surfaces in which disrupting digital infrastructure may cripple physical manufacturing despite its geographical distribution (Xu et al., 2018).

3.7. Balancing Multiple Risk Dimensions

My analysis shows that TSMC has truly contradictory risk objectives in which various risk reduction strategies generate natural tensions. Quick diversification would lower the risk of geopolitical concentration but raise the risk of strategic risk on loss of competitive advantage. Intellectual Property diffusion would run the risk of benefiting competitors as aggressive knowledge transfer would speed up the development of capabilities in Arizona. They are real

strategic dilemmas with no best answer, and not artificial complexity created to rationalize concentration.

TSMC needs to find actual trade-offs with various stakeholders having different priorities and focus on various risks depending on their positions and interests. The orientation of the company shows a complicated balancing of these conflicting pressures instead of straightforward strategic computation in the event of one goal. The convergence of real-life operational issues and strategic thinking situations results in the presence of cases, when a legitimate focus on the problems is conveniently and multi-purposely.

4. Performance System: Dual Metrics Framework

4.1. Operational vs. Strategic Performance

The performance measurement faced by TSMC has real opposing tensions among the evaluation frameworks that cater to the interest of various stakeholders. Operational efficiency is promoted with the help of traditional manufacturing metrics such as yield, cost per wafer, time-to-market (Kaplan & Norton, 1996). Strategic positioning metrics, such as technological leadership, sustainability of competitive advantage, dependency of customers, are concerned with long-term competitive positioning. At times, these frameworks contradict one another.

Table 2 and Table 3 compare the performance of TSMC in two different frameworks, namely the traditional operational metrics and the strategic positioning metrics. It is this two-fold framework that presents a paradox that is seemingly finely tuned Arizona is performing so poorly according to conventional metrics, that it is indicative of failure in execution. Nevertheless, when it comes to the aspect of strategic positioning, the operational problems in Arizona produce beneficial results as they show the real competitive edges of Taiwan.

Table 2 : Traditional Operational Performance Metrics: Taiwan vs. Arizona

Performance Metric	Taiwan Performance	Arizona Performance	Gap Analysis	Operational Assessment
Manufacturing Yield	90-92%	70-75% (3-5 year convergence target)	-17 percentage points	Arizona significantly underperforms
Cost per Wafer	$5,000 (baseline)	$6,500 (+30% premium)	+$1,500 cost disadvantage	Arizona fails cost competitiveness

Technology Generation	N (2nm in production)	N+2 (3nm target by 2026)	2 generations behind	Arizona perpetually lags cutting-edge
Local Sourcing %	100% (complete ecosystem)	30% (70% imported from Asia)	70% import dependency	Arizona lacks supply autonomy
Production Volume	20,000 wafers/month (target met)	10,000 wafers/month (50% reduction)	50% capacity shortfall	Arizona below planned capacity
Capacity Utilization	85-90% (optimized)	50-60% (ramp-up phase)	-30 percentage points	Arizona underutilizes investment
Time to Troubleshoot Issues	4-8 hours (local expertise)	24-48 hours (remote support needed)	3-6x slower resolution	Arizona operationally dependent

Table 3 Strategic Positioning Performance Metrics: Taiwan vs. Arizona

Strategic Metric	Taiwan Position	Arizona Position	Strategic Analysis	Positioning Assessment
Technology Leadership	Cutting-edge innovation (N, N+1 generation)	Follows with 1-2 generation lag	Taiwan maintains innovation primacy	Leadership concentration preserved
Customer Dependency Intensity	High (80%+ of volume allocations)	Limited (15-20% of volume)	Customers prefer Taiwan production	Strategic value validated by behavior
Supply Chain Autonomy	Complete ecosystem independence	70% Taiwan/Asia import reliance	Arizona structurally dependent	Ecosystem advantages demonstrated
Knowledge Concentration	Full proprietary knowledge base	25-30% tacit knowledge transferred	Expertise remains Taiwan-concent rated	Competitive advantage protected

Geopolitical Deterrence Value	High (Silicon Shield intact)	Partial (limited risk mitigation)	Taiwan's unique position preserved	Strategic irreplaceability maintained
Market Validation	$0 government subsidies required	$39B CHIPS Act subsidy required	True replication costs revealed	Arizona proves Taiwan's economic advantages
Talent Pipeline Control	2,000+ PhD engineers available locally	Depends on 500+ Taiwan rotations	Human capital advantages demonstrated	Taiwan maintains talent superiority

The benefits of performance are transformations into the leadership in innovation not only in direct financial terms. The 18-24 months head start of the N-generation process in Taiwan implies that engineers will have the experience of informing the next generation development. This accumulates with time: engineers who have mastered 2-nanometer process optimization develop an insight into 1.5-nanometer technologies currently in development, and Arizona engineers who have mastered 3-nanometer process optimization gain experience with mature technologies that are unlikely to provide them with future learning value (Nonaka and Takeuchi, 2019).

Financial results go beyond the operational margins to the strategic optionality. The 35-40% operating margins give Taiwan capital to invest in Taiwanese R&D of over $15 billion annually in keeping it technologically ahead. The 20-25% margins the state of Arizona is expected to receive cannot result in enough returns necessary to justify a 40-billion expenditure in the regular market, and thus it would be necessary to keep subsidizing it. This difference implies that the operations of Taiwan generate the innovation that sustains its advantage, whereas Arizona operations consume resources and create strategic asymmetry with Taiwan consolidating and distributed operations in general positioning (Kaplan and Norton, 1996).

4.2. Performance-to-Outcomes Causal Framework

The supply chain performance brings about strategic results in self-production processes. The high performance of Taiwan, which is 96/100 SCOR, 90%+ yields, contributes to the choices of customer allocation which focuses the volume on Taiwan. The 80%+ share allocation of Apple is a logical estimate: it is 17-percentage-point higher in Taiwan and 30-percent cost-benefit, and it results in a lower total cost and increased reliability (Bloomberg, 2024). This generates virtuous cycles whereby an increased amount of volume will lead to scale economies, worth further Taiwan investment, and top engineering talent- supporting the initial performance benefits.

The 36-point competitive advantage creates a performance gap that results in competitive advantage that is estimated to cost between $8-12 billion a year. High yields offer savings of around $2 billion; cost-effectiveness creates a benefit of $3 billion; consistency of quality can

offer a premium price of between 2-3 billion; the leadership in technology can create the first-mover advantage of between 3-4 billion (Fuller, 2022).

Table 4 Supply Chain Performance Linkage to Strategic Outcomes: Quantified Impact

Performance Metric	Taiwan	Arizona	Strategic Outcome	Quantified Impact
SCOR Score	96/100	60/100	Customer preference	80% vs 15-20% allocation
Yield	90-92%	70-75%	Cost advantage	~$2B annual benefit
Cost/Wafer	$5,000	$6,500	Margin differential	35-40% vs 20-25% margins
Technology Gen	N (2nm)	N+2 (3nm)	Innovation leadership	18-24 mo. first-mover; $3-4B premium
Local Sourcing	100%	30%	Supply flexibility	48-hr vs 2-3 week lead times
Volume	20K/month	10K/month	Market share	92% advanced logic dominance
Knowledge	Complete	25-30%	Competitive moat	3x easier talent acquisition
Overall	Superior	Inferior	**Taiwan irreplaceable**	**$8-12B annual advantage**

Table 4 measures the relationship between the 36-point SCOR gap and strategic performance: customer preference (80% Taiwan), financial performance (15-20 point margin difference), and

innovation leadership (18-24 month technology advantage). These form self-reinforcing loops of how excellent performance leads to volume, scale economies, investment in innovation, and performance superiority.

The 18-24 month head start used by Taiwan deploying N-generation processes is that engineers are able to acquire experience in informing next-generation development; a 2-nanometer process develops insight in a 1.5-nanometer technologies. Arizona engineers who have mastered 3-nanometer processes, two years later, have some experience with mature technologies that have little future value (Nonaka and Takeuchi, 2019). The 35-40 percent margins in Taiwan create 15+ billion R&D to continue to lead, and the 20-25 percent of Arizona cannot afford the 40 billion investment without subsidies- the operations in Taiwan create innovation and the ones in Arizona drain resources making strategic asymmetry (Kaplan and Norton, 1996).

4.3. Strategic Performance Indicators

In addition to more conventional metrics, TSMC has to evaluate strategic positioning dimensions:

Customer Dependency Intensity: 80%+ volume distribution to Taiwan even as Arizona is on board illustrates that Apple is being perceived by its customers that Taiwan production will provide real benefits (Bloomberg, 2024). That is pure judgment grounded on quality tests--though conveniently justifying the further strategic usefulness of Taiwan.

Technology Leadership Concentration: The trend of TSMC to introduce next-generation technologies in Taiwan, followed by Arizona, with a 1-2 generation gap, has been associated with both operational reasoning (introducing complicated processes in which resources are concentrated on the location) and strategic calculation (remaining at the technological edge in Taiwan).

Competitive Positioning: The competitive positioning of TSMC against the competitors is 96/100 SCOR against the estimated 75-85/100 of the competitors, which allows it to retain market share of 92 in advanced logic semiconductors (McKinsey & Company, 2022). The consequences of this dominance are not just in revenue: customers will create products optimized to the TSMC processes, which makes switching costly; the engineers will use TSMC to gain advanced experience; the suppliers will focus on TSMC relationships. These self-reinforcing processes make the performance of supply chains one of the competitive moats, which increase with time.

4.4. The Dual Assessment Framework in Arizona.

Arizona needs to identify operational measures as well as strategic contributions. Operationally, Arizona is far behind Taiwan in yield, cost and efficiency- which are real execution problems. At a strategic level, Arizona shows commitment to the diversification process, meets the

stakeholder needs, generates political goodwill, and offers evidence that the operation problems are real and impede fast replication of Taiwan.

This two-sidedness leads to a scenario in which the failure to perform in operations translates to strategic value, eliminating the need to take forceful action to correct the situation which can cause Arizona to grow faster than strategically comfortable development speeds. The 40billion investment is an indicator of true diversification seeking and constant dangers to Taiwan strategic position are constrained by its incessant problems.

Conclusion: Complexity, Authenticity, and Strategic Navigation

The underlying issue, which is discussed in this analysis, is whether the semiconductor manufacturing process can be distributed successfully and be efficient and innovative. The SCOR analysis, technology assessment, risk management evaluation, and performance metrics data allow one to conclude that the distribution is possible but limited in terms of qualified affirmative. The geographic distribution of manufacturing capacity can be made to do this - Arizona will make high-level chip production, but efficiency will be greatly impaired in the process. Taiwan (96/100) to Arizona (60/100) 36 point SCOR difference plus chronic cost advantages (30 plus) and yield losses (75 percent versus 90 percent and above) indicate that distributed operations cannot be as efficient as concentrated ecosystems in 5-10 year periods. Innovation is less accessible: high-technology process creation demands highly focused knowledge and ecosystems that are done in fragments by distribution. The solution proposed by TSMC, that is staging technology deployment in which Arizona is always several generations behind Taiwan, allows the distribution of capacities, but does not compromise the centralization of innovation. Therefore, the answer depends on how we define 'successful distribution.' If it means functioning facilities that produce advanced chips, the answer is yes Arizona will eventually manufacture semiconductors. However, if it means achieving Taiwan-equivalent efficiency and innovation leadership in distributed locations within policy-relevant planning horizons of 5-10 years, the answer is no. Authentic operational barriers require generational timeframes that policy cycles and market pressures cannot accommodate.

The geographic expansion of TSMC demonstrates the presence of how real-life challenges of supply chains can match the strategic factor in the manner that ultimate intentions are hard to determine. Some of the genuine operational constraints affecting the company include; ecosystem dependencies that take decades to be fully developed, tacit knowledge that cannot be codified and transferred across environments, cultural differences that hinder cross-cultural organizational learning, and the complexity involved in coordination that accompanies operations that are geographically distributed. These are actual challenges, which have been substantiated by a significant amount of scholarly investigation on industrial cluster, knowledge transfer, and manufacturing geography.

But these real challenges happen to be of strategic value in terms of maintaining competitive advantages of Taiwan and manifesting the reality that it is difficult to develop Taiwan substitutes practically within the timeframes relevant to policy. My SCOR analysis reveals how concentration of supply chain processes in Taiwan are focused on genuine operations reasons

and at the same time offers strategic positioning advantages. The technology analysis indicates that there are real weaknesses in technology-mediated knowledge transfer besides the fact that these weaknesses maintain the human expertise benefits that can hardly be matched by the competitors.

The risk management analysis reveals the actual tensions among conflicting strategic objectives where one mitigation approach involves an in-built trade-off. Performance measurement systems unveil the conflict between operational measures and strategic positioning measures in some cases, to the effect that poor performance in terms of conventional efficiency measures, ironically, creates strategic value in terms of manifested indispensability.

In my analysis, I put much emphasis on the fact that TSMC might be ambivalent about the overall success of diversification, instead of being strategic resistance to expansion in geographic directions. The company is advantaged by keeping Taiwan at the center yet it actually requires fulfilling the demands of the stakeholders to have choices of supply diversification. This presents complicated scenarios as TSMC embarks on diversification with good intention and heavy capital outlay hoping that the results will work out just well enough to meet the demands of the immediate stakeholders without going so far as to completely remove Taiwan of its strategic niche.

The overlap of genuine operational issues and strategic issues implies that TSMC is able to maneuver these tensions and retain plausible deniability regarding the eventual motives. The company is not obliged to intentionally undermine the operations in Arizona or intentionally restrict the development, there are actual impediments that restrain the development and TSMC maintains discretion on the extent to which the impediments may be overcome by the allocation of resources and organizational priorities.

This analysis shows TSMC's expansion isn't just about meeting customer demands or putting on a show. The truth is messier. Real supply chain problems end up creating strategic advantages, almost as a side effect. That's more useful to understand than trying to figure out if TSMC is deliberately dragging its feet or just dealing with a really tough situation, one where different stakeholders want different things and there's no perfect answer.

Take the Arizona facility. It scores 36 points lower than Taiwan on supply chain metrics. Part of that gap is real Taiwan has decades of experience Arizona doesn't have yet. But it also works out nicely for keeping Taiwan ahead in the competitive race. That's basically how TSMC handles its whole geographic expansion: operational reality and strategic positioning blend together.

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